WFLOW-04: Advanced Notification Routing

> **Series:** WFLOW — Workflows and Alert Notifications | **Notebook:** 4 of 10 | **Created:** January 2026 | **Last Updated:** 10/23/2026

Intelligent Alert Routing

Not all alerts should go to everyone. This notebook covers conditional routing based on severity, team ownership, time of day, and escalation patterns.

Table of Contents

1. Routing Strategies
2. Conditional Expressions
3. Routing by Severity
4. Routing by Team/Service
5. Time-Based Routing
6. Escalation Patterns
7. Multi-Channel Strategy

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription

Requirement	Details
Permissions	`automation:workflows:write` ; the workflow actor also needs `storage:events:read` for the Problem trigger
Prior Knowledge	**WFLOW-01** through **WFLOW-03**
Connections	Slack and/or Teams connections configured

Sprint 1.337 (April 2026): Smartscape Ownership for Routing

SaaS 1.337 made ownership available on Smartscape nodes, for use from **Workflows actions**: *"Ownership information is now available in Smartscape and for Smartscape nodes. This lets you use Workflows actions (such as get_owners) to send automated and targeted notifications to your teams"* (What's new in SaaS 1.337 (DT docs)). It is not a DQL field. An earlier revision of this notebook read `getNodeField(affected_entity_ids[0], "ownership.team")` ; that returns null on every problem, because no Smartscape node model has an `ownership.*` field and `getNodeField` does not resolve a classic entity ID.

Two documented routes:

- **Route on ownership with the Ownership app's *Get owners* action.** Add a *Get owners* task after the Problem trigger. Its *Entity ids* input takes *"A Jinja expression, for example,* `{{ event()["dt.entity.host"] }}` *"*, but that field is rarely set on a problem (33 of 4,244 problems over 7 days on the validation tenant, 10/06/2026). Pass the problem's entities instead, `{{ event()["affected_entity_ids"] | join(",") }}` ; the input takes several: *"Use a comma or semicolon to separate multiple IDs."* The output carries `slackChannels` , `msTeams` , `email` and `jira` , and each is a list per team (*"A list of Slack contact details per team"*), not a single channel (Actions for Ownership (DT docs)). Loop the Slack task over `result("get_owners").slackChannels` with the **Loop task** option and take the channel from each item; run the workflow once and read an item's shape in the Results tab before wiring it. That replaces hard-coded team names in the workflow.
- **Or route on** `primary_tags.team` in the trigger's custom filter or a task condition. *"Dynatrace enriches all derived signals (service metrics, Davis events, and problems) with the same tags."* (Primary Grail fields and tags (DT docs)). The matcher is `primary_tags.team == "payments"` . Verify your entities carry the tag first: on the validation tenant `primary_tags.team` was null on every problem (09/24/2026).

FAQ-21 §5 covers ownership coverage and the Problem fields mapping in depth.

Dynatrace API 1.337: `metadata` removed from the Environment API v1 events endpoints

The Dynatrace API 1.337 changelog lists, under **Environment API v1**, `GET /events` and `GET /events/{eventId}` with *"Changed EventQueryResult schema (application/json) Changed property events Removed properties: metadata"*. This affects only an HTTP task that calls the v1 endpoints `/api/v1/events` or `/api/v1/events/{eventId}` and parses `metadata`; drop that reference. Workflows started by a Problem or Davis event trigger read Grail records, which never had that property, so they are unaffected.

> **Sources:** Dynatrace API changelog 1.337 (DT docs).

---

1. Routing Strategies

> ⚠️ **Migrating from alerting profiles? Duration suppression maps onto the trigger's Minimum duration option.** An alerting profile can delay notification until a problem has been open longer than *N* minutes (`delayInMinutes`) — a common way to suppress transient blips. The problem trigger's **Minimum duration** option (renamed from **Delay** in 08/2026) postpones *"the trigger until the problem has been open for at least the configured duration"* — 5, 10, 15, 30, 60, 120, 240, 1440, or 10080 minutes, evaluated on `dt.duration_marker` (Problem and event triggers (DT docs)). Since its 09/07/2026 rewrite the alert-notification upgrade guide (DT docs) agrees: *"The delay, update, and severity capabilities described in this guide exist only on the workflow trigger."* Earlier versions of that guide said there was no alternative; that line is gone.
>
> The values are fixed, so a delay that is not one of those nine has to be rounded — up suppresses a little more, down pages a little sooner. Treat a long delay as evidence the alert was the wrong *shape* rather than merely delayed: a profile suppressing 30 minutes of a firing condition is usually describing a burn-rate concern, which belongs in an SLO burn-rate alert (**SLO-04**) or a Davis anomaly detector (**AIOPS-02**). Inventory `delayInMinutes` across every profile and choose a Minimum duration for each; no special migration wave is needed. See **MZ2POL-09** §6.1.

Why Route Alerts?

Problem	Impact	Solution
Alert fatigue	Teams ignore alerts	Route only relevant alerts
Slow response	Wrong team notified	Route to owners
Off-hours noise	Sleep disruption	Time-based routing
Missed escalation	Unacknowledged alerts	Auto-escalation

Routing Dimensions

Dimension	Route Based On	Example
Severity	`event.severity` (1–5)	Critical (1) → PagerDuty, Minor (3) → Slack
Team	`owner` / `dt.owner` tags, or Smartscape ownership	`owner:checkout` → #checkout-alerts
Service	Service name or entity ID	Payment service → payments team
Time	Hour of day, day of week	Weekends → on-call only
Environment	prod/staging/dev tags	Prod → immediate, Dev → daily digest

2. Conditional Expressions

A task condition decides whether a task runs. There is no workflow-level list of named conditions: each task carries **one** condition, written under the task itself. *"You can express task conditions based on the final state of the predecessor task and as a custom condition to implement any custom logic."*

Part	YAML key	What it holds
State condition	`states`	One entry per predecessor task. The editor offers *success or skipped* (the default), *success*, *error or cancelled*, *error* and *any*; exported YAML uses `OK`, `SUCCESS`, `NOK`, `ERROR` and `ANY`, and Dynatrace's samples use `OK`
Custom condition	`custom`	**One** Jinja expression; the task runs only when it is true. Combine checks with `and` / `or` / `not` inside it
Else	`else`	`STOP` or `SKIP`. The default is *stop here*: *"if you want to skip this step and continue the workflow, you can change the else action from stop here (default) to skip"*

A task that follows the trigger directly has no predecessor: *"It's not possible to add a Run this task if and Else condition to the first task."* It can still carry the custom condition: *"Nevertheless, a custom condition can be configured to evaluate the properties of an event trigger context."* Dynatrace's own multi-channel sample nonetheless sets `else: SKIP` on its parallel first tasks in YAML, and the examples in this notebook follow it. In a routing workflow, set Else to **Skip** on every branch that may legitimately not run.

Connection inputs take a connection **ID** (Slack: *"Connection ID."*). In YAML, resolve a connection by its name with `connection()`: *"Get a single connection by schema ID and connection name."* The connection picker in the editor fills the ID for you.

A Condition on a Task

```
tasks:
  pagerduty_alert:
    name: pagerduty_alert
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      # One expression: AND, OR and NOT go inside it
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      # ... event inputs as in §3
```

Negating a Condition

```
tasks:
  slack_only_for_non_critical:
    name: slack_only_for_non_critical
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ not ((event().get("event.severity") | int(5)) &lt;= 1) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: "{{ event()['event.name'] }}"
```

Conditioning on a Predecessor's State

```
tasks:
  notify_on_failure:
    name: notify_on_failure
    action: dynatrace.slack:slack-send-message
    predecessors:
      - create_incident
    conditions:
      states:
        create_incident: ERROR    # run only if create_incident failed
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-integration"
      message: "Incident creation failed for {{ event()['display_id'] }}"
```

Expressions

```
# AND within one expression
{{ (event().get("event.severity") | int(5)) &lt;= 1 and "env:prod" in
(event().get("entity_tags") or []) }}

# Critical or Major (1 or 2)
{{ (event().get("event.severity") | int(5)) &lt;= 2 }}

# Check if field exists
{{ event().get("root_cause_entity_id") is not none }}
```

Read any field a problem may lack with `.get()` and a default. The reference documents that reading a missing attribute fails with an *"Undefined variables"* message (shown there for `input()` and `result()`), so the task errors rather than evaluating false. Problem records carry tags in `entity_tags`, an array of `key:value` strings, so `in` tests for an exact element. The field is deprecated (*"`entity_tags` still exists but is deprecated (Dynatrace Classic)"*): prefer the trigger's tag filter (§4) or `primary_tags.*`. An earlier `tags` string field stopped appearing on problem records on 10/08/2026 on the validation tenant, so a condition that reads `tags` is always false and its task skips.

> **Sources:**
>
> - Build workflows (DT docs)
> - Jinja expressions for Workflows (DT docs)
> - Upgrade from Classic problem notification to simple workflows (DT docs) — maps `{Tags}` to *"primary_tags.* entity_tags still exists but is deprecated (Dynatrace Classic)"*
> - **Dictionary:** model `davis.problem` (`dt.davis.problems`) lists `entity_tags` and no `tags` field, read 10/15/2026. Tenant: `countIf(isNotNull(tags))` 0 of 4,282 problems over 7 days, `entity_tags` (array) on 2,076.
> - Slack Connector actions (DT docs)
> - threat-detection-notification-sender.yaml (Dynatrace GitHub) — parallel first tasks with `states: {}` , a `custom` condition and `else: SKIP`

3. Routing by Severity

Severity-Based Routing Pattern

The problem record carries severity as `event.severity`: *"Severity is expressed as a numeric value from 1 (the most critical state) to 5 (purely informational event)."* The levels are named 1 Critical, 2 Major, 3 Minor, 4 Warning and 5 Informational. There is no `CRITICAL` / `HIGH` / `MEDIUM` / `LOW` string on the record, so conditions comparing against those strings are false for every problem.

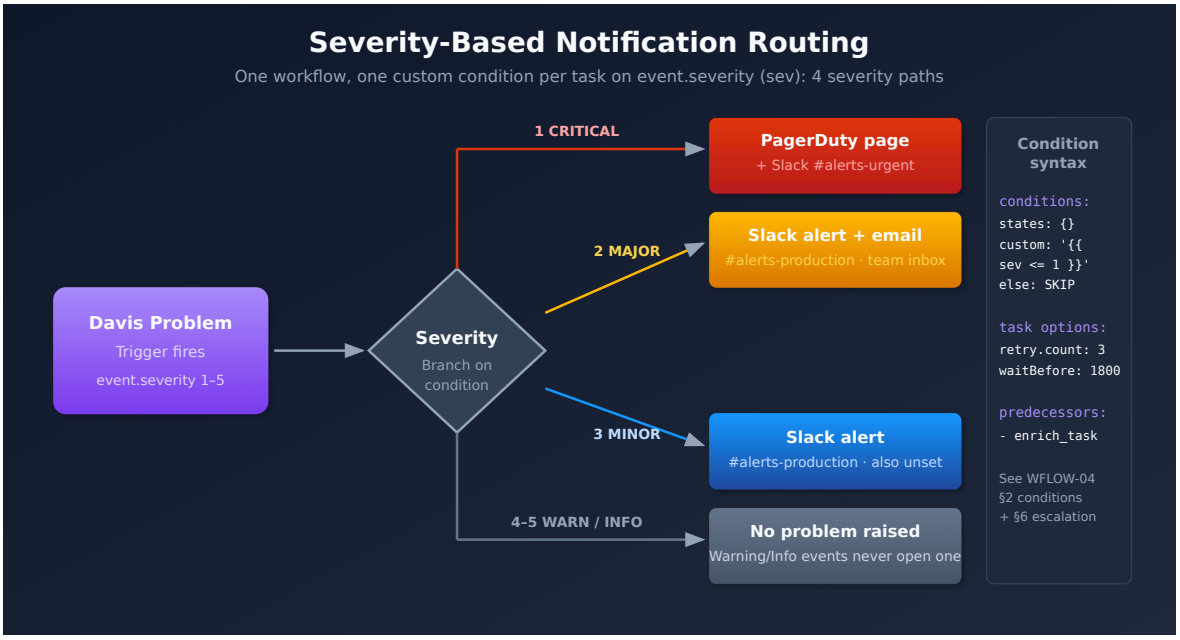
Problems carry only the first three levels. *"The problem inherits the highest (most critical) severity across all grouped events." "Warning and Informational events appear in the event list and can be grouped into a problem by problem correlation, but they don't raise problems on their own."* On the validation tenant, 3,755 of 4,244 problems in the 7 days to 10/06/2026 carried `3` and the other 489 carried `1`; none carried `2`, `4` or `5`. Route on the tiers below:

`event.severity`	**Actions**
1 (Critical)	PagerDuty + Slack #alerts-urgent
2 (Major)	Slack #alerts-production + email to the team
3 (Minor), or unset	Slack #alerts-production

To drop low tiers before any condition runs, set the Problem trigger's **Severity** filter: *"Selecting a severity level triggers the workflow for that level and all higher (more critical) levels."*

> **Breaking — SaaS 1.348 (rollout start 10/12/2026, staged tenant rollout): severity is no longer defaulted.** *"Davis events and problems no longer default `event.severity` to `3`."* A severity filter that was matching the default stops matching once 1.348 reaches your tenant: *"Workflows with a Davis event/problem trigger that filter on `event.severity=3` expecting it to be defaulted, might need to be changed to filter for `Any` severity to keep being triggered."* Before relying on severity tiers, check which event sources actually **set** a severity. For sources that do not, route on `event.category`, ownership or tags instead, or restore a default with a Davis event OpenPipeline processor. Until 1.348 reaches your tenant, the defaulting behavior still applies and the conditions below keep working. The `int(5)` in them

> treats a missing severity as the least severe tier, so an unset severity lands with Minor and never pages. **FAQ-21 §6** carries the same note.
>
> **Sources:** Standardized event severity (DT docs), What's new in Dynatrace SaaS 1.348 (DT docs) — GA, rollout start 10/12/2026; re-read 10/23/2026. **Dictionary:** `event.severity` (`experimental`), read 10/06/2026.



Workflow Configuration

```
tasks:
  # 1 Critical: page on-call + Slack urgent
  pagerduty_critical:
    name: pagerduty_critical
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"   # an Events connection (routing key)
      eventAction: trigger
      severity: critical
      summary: "{{ event()['event.name'] }}"
      source: dynatrace
      dedupKey: "dynatrace-{{ event()['display_id'] }}"

  slack_urgent:
    name: slack_urgent
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-urgent"
      message: ":rotating_light: *CRITICAL* {{ event()['event.name'] }}"

  # 2 Major: Slack alerts (add an email task with the same condition)
  slack_major:
    name: slack_major
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) == 2 }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
```

```
      channel: "#alerts-production"
      message: ":warning: *MAJOR* {{ event()['event.name'] }}"

  # 3 Minor, or unset: Slack only
  slack_minor:
    name: slack_minor
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &gt;= 3 }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: ":information_source: *{{ event()['event.category'] }}* {{
event()['event.name'] }}"
```

Severity levels. `event.severity` is an integer from 1 to 5, with the names Dynatrace uses in the Problem trigger and the problems list:

Integer	Level
1	Critical
2	Major
3	Minor
4	Warning
5	Informational

The conditions in this section key on this numeric field. Earlier revisions compared against `"CRITICAL"` / `"HIGH"` / `"MEDIUM"` / `"LOW"`; the problem record carries no such strings, so those conditions were false for every problem. *"Grail stores severity as a numeric value."* The conditions still convert with `| int`, because the value in the `event()` payload may arrive as a string; Dynatrace's own PagerDuty template likewise normalizes it with `| string | trim` before mapping. The field is `experimental` in the semantic dictionary. In DQL (problem feed, reporting) it is queried directly as `event.severity`; see **AIOPS-03 §5** for the severity rollup pattern.

> **Sources:** Standardized event severity (DT docs), Send problem as event to PagerDuty template (DT docs). **Dictionary:** `event.severity` (`experimental`, type `long`), read 10/06/2026.

4. Routing by Team/Service

Route by Ownership Tags (recommended)

Dynatrace's ownership keys are built in: *"By default, Dynatrace provides two keys: owner and dt.owner ."* Tag entities with them, then give each team a workflow whose Problem trigger filters on its tag with the trigger's *"Affected entities : Filter by entity tags"* option. The routing decision is then made before any task runs, and no condition is needed. **FAQ-21 §5** walks through this pattern.

Tag Conditions in One Workflow (fallback)

When one workflow must fan out to several teams, test the problem's tags in each task's custom condition. `entity_tags` is an array, so these are exact matches (§2).

```
tasks:
  notify_checkout:
    name: notify_checkout
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ "owner:checkout" in (event().get("entity_tags") or []) or
"owner:cart" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#checkout-alerts"
      message: "{{ event()['event.name'] }}"

  notify_payments:
    name: notify_payments
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ "owner:payments" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#payments-alerts"
      message: "{{ event()['event.name'] }}"

  notify_platform:
    name: notify_platform
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ "owner:platform" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#platform-alerts"
      message: "{{ event()['event.name'] }}"
```

> **Sources:** Assign team ownership (DT docs), Problem and event triggers (DT docs).

Management Zone Routing (legacy — do not build new workflows on this)

> ⚠️ **This pattern has nothing to match.** `management_zones` is **not present on problem records**: on a live tenant (10/05/2026) 0 of 15,878 problems in `dt.davis.problems` over 30 days carried it, while 14 of 15 hosts belonged to a management zone. A condition that reads it has nothing to evaluate against, whether or not your zones still exist. There is no Management Zone filter on the problem trigger itself (WFLOW-02 §2).
>
> Use the ownership-tag pattern above instead. If a region is the routing dimension, tag the entities (`region:us-east`) or use Smartscape ownership rather than reading the MZ array. Teams actively migrating off MZs should read MZ2POL-01 §5.

To confirm on your own tenant: `fetch dt.davis.problems, from:-30d | summarize n = countIf(isNotNull(management_zones))` .

Dynamic Channel Selection

Use JavaScript to determine the channel dynamically:

```
import { execution } from '@dynatrace-sdk/automation-utils';

export default async function () {
  const ev = (await execution()).params.event;   // trigger payload
  // entity_tags still works but is deprecated (Dynatrace Classic) - see the
upgrade guide
  const tags = ev.entity_tags || [];            // "key:value" strings

  // Find the ownership tag
  const ownerTag = tags.find(t =&gt; t.startsWith('owner:'));
  const team = ownerTag ? ownerTag.split(':')[1] : 'platform';

  return {
    channel: `#${team}-alerts`,
    team: team
  };
}
```

> **Sources:** Upgrade guide — alert notification (DT docs) — *"entity_tags still exists but is deprecated (Dynatrace Classic)"*.

5. Time-Based Routing

Business Hours vs Off-Hours

`now()` is UTC unless you pass a time zone: *"If no timezone is provided, UTC is used."* Without one, 09:00–17:00 below would mean 09:00–17:00 UTC, which is 05:00–13:00 in New York, and the weekday boundary would fall at midnight UTC. Pass the team's zone (a pytz name) to every time check:

```
tasks:
  # Business hours (New York): Slack channel
  slack_business_hours:
    name: slack_business_hours
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ now("America/New_York").weekday() &lt; 5 and
now("America/New_York").hour &gt;= 9 and now("America/New_York").hour &lt; 17
}}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: "{{ event()['event.name'] }}"

  # Off-hours: PagerDuty for Critical only
  pagerduty_off_hours:
    name: pagerduty_off_hours
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (now("America/New_York").weekday() &gt;= 5 or
now("America/New_York").hour &lt; 9 or now("America/New_York").hour &gt;= 17)
and (event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      eventAction: trigger
      severity: critical
      summary: "[Off-Hours] {{ event()['event.name'] }}"
      source: dynatrace
      dedupKey: "dynatrace-{{ event()['display_id'] }}"

  # Off-hours, not Critical: queue for morning
  slack_queue:
    name: slack_queue
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (now("America/New_York").weekday() &gt;= 5 or
now("America/New_York").hour &lt; 9 or now("America/New_York").hour &gt;= 17)
```

```
and not ((event().get("event.severity") | int(5)) &lt;= 1) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-queue"
      message: ":moon: *Queued for review* {{ event()['event.name'] }}"
```

For holidays and irregular hours, the same reference documents business calendars, read with `calendars()`.

Weekend-Only Routing

```
tasks:
  weekend_oncall:
    name: weekend_oncall
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ now("America/New_York").weekday() &gt;= 5 and
(event().get("event.severity") | int(5)) &lt;= 1 }}'
      else: SKIP
    input:
      # Page the weekend on-call rotation: a connection that holds that
rotation's
      # routing key. Credentials live in connections, not in expressions.
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-weekend-oncall') }}"
      # ... event inputs as in §3
```

> **Sources:** Jinja expressions for Workflows (DT docs) — *"The current timestamp in UTC or time zone is provided as the input parameter."*

6. Escalation Patterns

Timed Escalation

Escalate if the problem is still active after a time window. This pattern checks the Dynatrace problem's `event.status`; it does not see whether anyone acknowledged the

alert in Slack or PagerDuty. In community practice, acknowledgment-aware escalation is left to the paging tool's own escalation policy.

```
tasks:
  # Step 1: Initial notification
  initial_slack:
    name: initial_slack
    action: dynatrace.slack:slack-send-message
    predecessors: []
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-production"
      message: ":warning: {{ event()['event.name'] }} - paging on-call if
still active in 15 min"

  # Step 2: After 15 minutes, check if still open (event.status is ACTIVE or
  # CLOSED — never OPEN). There is no "wait" action: the delay is this task's
  # own Wait before option, in seconds (max 86400).
  check_problem_status:
    name: check_problem_status
    action: dynatrace.automations:run-javascript
    predecessors:
      - initial_slack
    waitBefore: 900
    input:
      script: |
        import { execution } from '@dynatrace-sdk/automation-utils';
        import { queryExecutionClient } from '@dynatrace-sdk/client-query';

        export default async function () {
          const ev = (await execution()).params.event;   // snapshot from
trigger time
          // Re-query: the trigger payload is 15 minutes old by now
          const started = await queryExecutionClient.queryExecute({ body: {
requestTimeoutMilliseconds: 30000, query:
            `fetch dt.davis.problems, from:-2h | filter display_id ==
"${ev['display_id']}" | sort timestamp desc | limit 1 | fields event.status` }
});
          let q = started;   // no result yet if the query outlasted the
timeout - poll (see WFLOW-08 §2)
          while (q &amp;amp;&amp;amp; (q.state === 'NOT_STARTED' || q.state ===
'RUNNING')) {
            q = await queryExecutionClient.queryPoll({ requestToken:
started.requestToken, requestTimeoutMilliseconds: 30000 });
          }
          if (!q || q.state !== 'SUCCEEDED') throw new Error(`DQL query did
not succeed: ${q?.state ?? 'no response'}`);
          return { escalate: q.result.records[0]?.['event.status'] ===
'ACTIVE' };
```

```
        }

  # Step 3: Escalate to PagerDuty
  escalate_pagerduty:
    name: escalate_pagerduty
    action: dynatrace.pagerduty:send-event
    predecessors:
      - check_problem_status
    conditions:
      states:
        check_problem_status: OK
      custom: '{{ result("check_problem_status").escalate }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      eventAction: trigger
      severity: error                 # Events API v2 accepts critical,
error, warning, info
      source: dynatrace
      dedupKey: "dynatrace-{{ event()['display_id'] }}"
      summary: "[ESCALATED] {{ event()['event.name'] }} - still active after
15 min"
```

Workflows has no separate wait action. **Wait before** is a task option: *"The Wait before option controls how long a task stays in the waiting state before being run."* It takes seconds, up to 86,400, or a Jinja expression.

The same effect needs no waiting task at all: create a second workflow whose Problem trigger has **Minimum duration** 15 minutes and holds only the PagerDuty task. *"Minimum duration : Postpones the trigger until the problem has been open for at least the configured duration."* A problem that closes within 15 minutes never starts it.

> **Sources:** Build workflows — Task options (DT docs) — *"The default Wait is 0 second and max value is 86400 seconds."*; PagerDuty Connector actions (DT docs) — *"Trigger, acknowledge, or resolve an alert in PagerDuty using the Events API v2"*; Problem and event triggers (DT docs).

Multi-Tier Escalation

```
0 min  → Slack channel notification
15 min → Email to team lead
30 min → PagerDuty to on-call
60 min → PagerDuty to manager
```

7. Multi-Channel Strategy

Recommended Channel Matrix

`event.severity`	Environment	Channel(s)
1 (Critical)	Production	PagerDuty + Slack urgent + Email
1 (Critical)	Staging	Slack alerts
2 (Major)	Production	Slack alerts + Email
2 (Major)	Staging	Slack alerts
3 (Minor)	Production	Slack alerts
3 (Minor)	Staging	Slack (daily digest)
unset (SaaS 1.348+)	Any	Slack (daily digest), until the source sets a severity

Levels 4 (Warning) and 5 (Informational) do not raise problems on their own (§3), so a problem-triggered workflow needs no row for them.

Complete Multi-Channel Workflow

```
tasks:
  # PagerDuty: Critical + Production
  pagerduty:
    name: pagerduty
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-prod') }}"
      # ... event inputs as in §3

  # Slack Urgent: Critical + Production
  slack_urgent:
    name: slack_urgent
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
      channel: "#alerts-urgent"
      message: ":rotating_light: {{ event()['event.name'] }}"

  # Slack Standard: Major or above + Production
  slack_standard:
    name: slack_standard
    action: dynatrace.slack:slack-send-message
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 2 and
"env:prod" in (event().get("entity_tags") or []) }}'
      else: SKIP
    input:
      connection: "{{ connection('app:dynatrace.slack:connection', 'slack-
production') }}"
```

```
      channel: "#alerts-production"
      message: "{{ event()['event.name'] }}"

  # Email: Critical, or Major + Production
  email_alert:
    name: email_alert
    action: dynatrace.email:send-email
    predecessors: []
    conditions:
      states: {}
      custom: '{{ (event().get("event.severity") | int(5)) &lt;= 1 or
((event().get("event.severity") | int(5)) == 2 and "env:prod" in
(event().get("entity_tags") or [])) }}'
      else: SKIP
    # input: to, subject, content
```

Query Workflow Routing Effectiveness

```
// Workflow executions by outcome
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`,
and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in
any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to
`dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by
`event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION ·
WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript,
send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR ·
DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it
is ERROR
//   .state.is_final                    true only on terminal records —
filter on it, otherwise a
//                                        single execution is counted once as
RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                        (was task.error)  ·  duration (was
task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT"
| limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "WORKFLOW_EXECUTION"
| filter dt.automation_engine.state.is_final == true
| summarize executions = count(), by:{dt.automation_engine.state}
| sort executions desc
```

```
// Task execution distribution by notification channel
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`, and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to `dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by `event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION · WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript, send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR · DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it is ERROR
//   .state.is_final                     true only on terminal records — filter on it, otherwise a
//                                        single execution is counted once as RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                         (was task.error)  ·  duration (was task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT" | limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "ACTION_EXECUTION"
| filter dt.automation_engine.state.is_final == true
| summarize executions = count(), by:{dt.automation_engine.action.app, dt.automation_engine.action.function}
| sort executions desc
| limit 20
```

```
// Skipped tasks (conditions not met)
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`, and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to `dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by `event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION · WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript, send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR · DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it is ERROR
//   .state.is_final                     true only on terminal records — filter on it, otherwise a
//                                        single execution is counted once as RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                         (was task.error)  ·  duration (was task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT" | limit 1
fetch dt.system.events, from:-24h
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "TASK_EXECUTION"
| filter in(dt.automation_engine.state, {"SKIPPED", "DISCARDED"})
| summarize skipped = count(), by:{dt.automation_engine.workflow.title, dt.automation_engine.task.name, dt.automation_engine.state}
| sort skipped desc
| limit 25
```

Next Steps

With routing configured, integrate with incident management:

Recommended Path

1. **WFLOW-05: PagerDuty & ServiceNow** - Create incidents automatically
2. **WFLOW-06: Custom Templates** - Rich message formatting
3. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation

Key Takeaways

- **Conditions** live on each task: predecessor states, one custom Jinja expression, and an Else of Stop or Skip
- **Severity routing** keys on `event.severity` 1 Critical / 2 Major / 3 Minor; problems never carry 4 or 5
- **Team routing** uses `owner` / `dt.owner` tags (best in the trigger's tag filter) or Smartscape ownership — not management zones
- **Time-based routing** reduces off-hours noise; pass a time zone to `now()`, which is UTC otherwise
- **Escalation patterns** prevent missed alerts
- Test conditions with On-Demand trigger before production

Summary

In this notebook, you learned:

- Why and when to route alerts conditionally
- Conditional expression syntax and operators
- Severity-based routing patterns
- Team routing by entity tag, and why management-zone routing is now legacy
- Time-based (business hours) routing
- Escalation with wait-and-check patterns, or a second workflow with a Minimum duration
- Multi-channel notification strategy

References

- Workflow reference / Jinja expressions + conditions (DT docs)
- Build workflows — task conditions and options (DT docs)
- Problem and event triggers (DT docs)
- Standardized event severity (DT docs)
- Workflow actions umbrella (DT docs)
- Notification actions umbrella (DT docs)
- Davis Problems app (DT docs)
- Alerting and notifications umbrella (DT docs)
- Upgrade guide — alerting and notifications (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.